PCB meta analysis data to Atlantis format

2024-12-18

Functions defining from Raisha paper (Lovindeer et al. in prep)

Equation 10

```
M_contam_daily <- function(C_bio, LC50, beta_LC){
  return(log(1+(C_bio/LC50)^beta_LC)/86400)
}
```

Equation 11

```
k_growth <- function(C_bio, EC50, beta_EC){
  return(1/(1+C_bio/EC50)^beta_EC)
}
```

Equation 12

```
k_growth2 <- function(C_bio, k, c_range, c_mid){
  return(1-k/(1+exp(c_range*(c_mid-C_bio))))
}
```

Equation 13

```
k_repro <- function(C_bio, w_r, c_chronic){
  return(1-(w_r*(C_bio-c_chronic)))
}
```

Import data from Berninger & Tillitt 2018 (Table 5)

Paper DOI: https://doi.org/10.1002/etc.4335

Row data

```r
# Table SI 8
data_regression <- read.csv(here("Data/Meta_analysis_PCB_effects_on_fish_regression.csv"))
concentration = 10^(seq(-4,5,0.01))

# % reduction growth
growth = data_regression$intercept[3] + data_regression$slope[3]*log10(concentration)
growth[growth<0]<-0
growth[growth>100]<-100

# % reduction reproduction
reproduction = data_regression$intercept[5] + data_regression$slope[5]*log10(concentration)
reproduction[reproduction<0]<-0
reproduction[reproduction>100]<-100

# % population facing mortality
mortality = data_regression$intercept[1] + data_regression$slope[1]*log10(concentration)
mortality[mortality<0]<-0
mortality[mortality>100]<-100

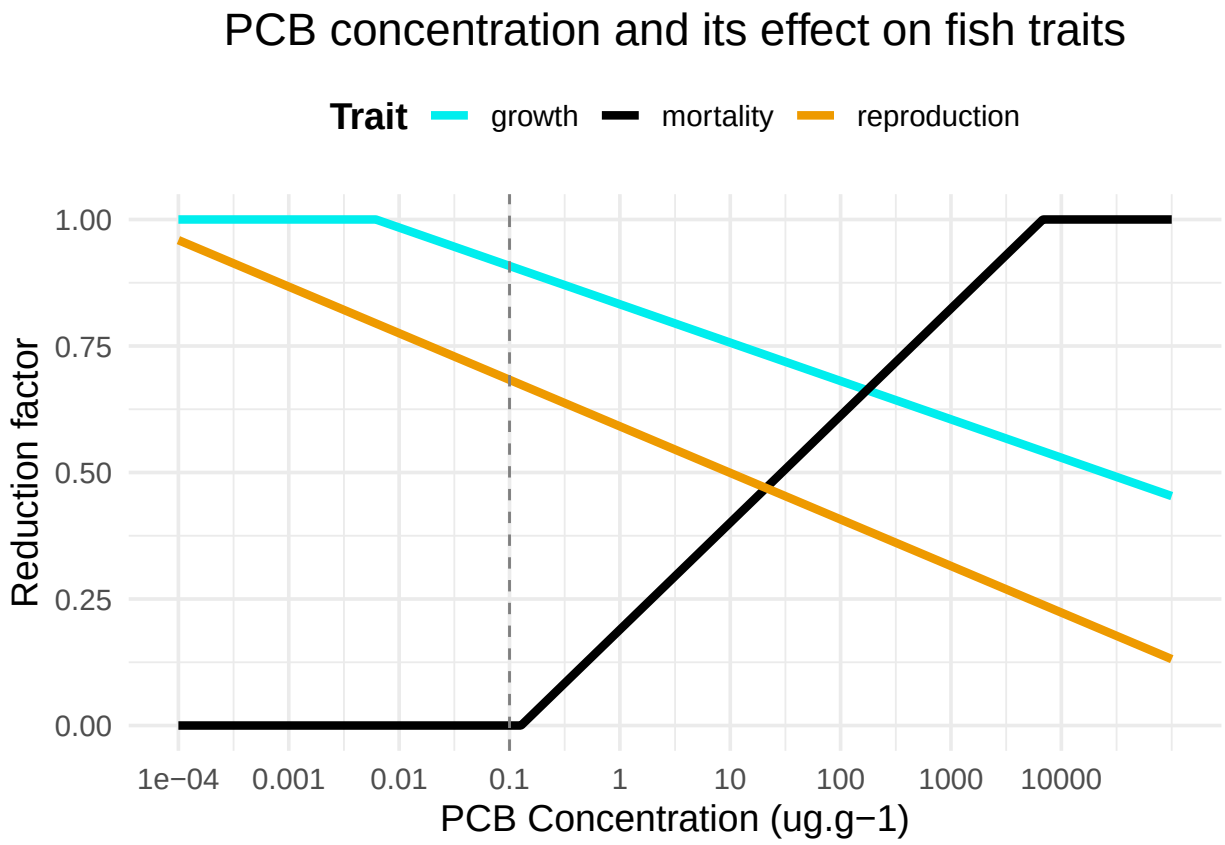

dataset <- data.frame(growth = 1-growth/100,
                      reproduction = 1-reproduction/100,
                      mortality = mortality/100,  # ATTENTION: it is not a rate
                      Concentration = concentration)


data_long <- dataset %>%
  tidyr::pivot_longer(cols = c(mortality, reproduction, growth),
                      names_to = "Trait",
                      values_to = "Value")


ggplot(data_long, aes(x = log10(Concentration), y = Value, color = Trait)) +
  geom_line(size = 1.5) +
  geom_vline(xintercept = -1, linetype = "dashed", color = "gray50") +
  scale_color_manual(values = c("mortality" = "black", "growth" = "cyan2", "reproduction" = "orange2")) +
  labs(    x = "PCB Concentration (ug.g-1)",
    y = "Reduction factor",
    color = "Trait",
    title = "PCB concentration and its effect on fish traits"  ) +
   scale_x_continuous(breaks=c(-4:4), labels=c(0.0001,0.001,0.01,0.1,1,10,100,1000, 10000)) +

  theme_minimal(base_size = 14) +
  theme(legend.position = "top",
    legend.title = element_text(face = "bold"),
    plot.title = element_text(hjust = 0.5)  )
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

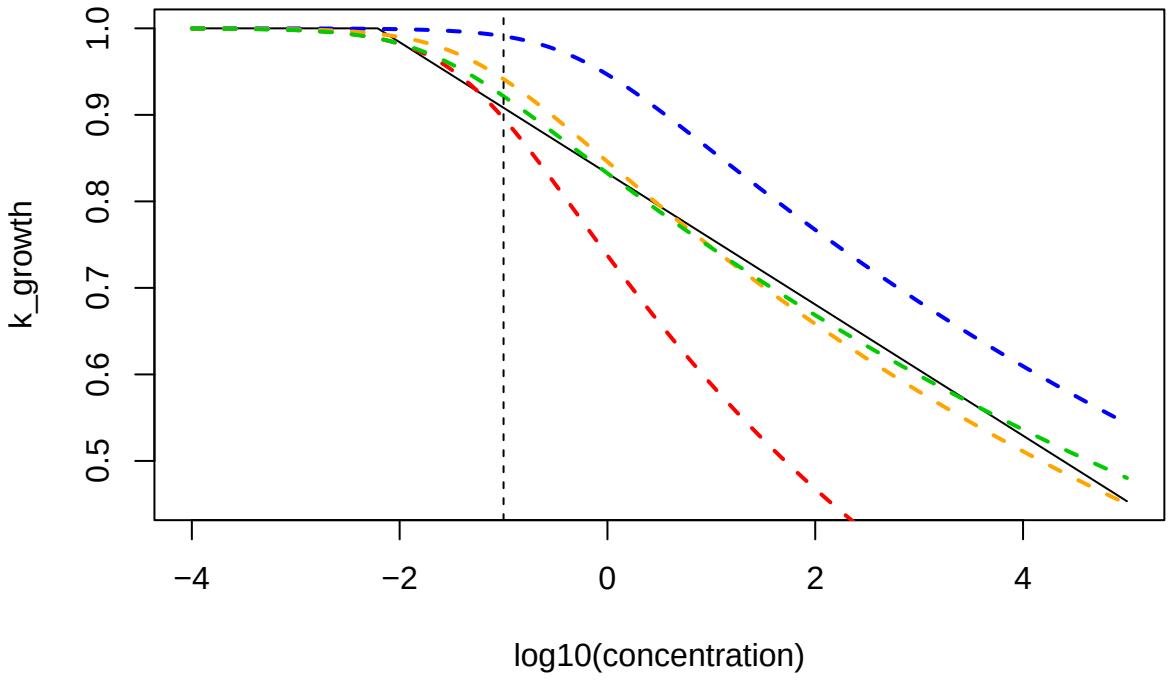


Growth

Fit equation 11 to data

```
{plot(log10(concentration), 1-growth/100, ylab = "k_growth", type = "l")
lines(log10(concentration), k_growth(concentration, 0.05, 0.1), lty = 2, col = "red", lwd = 2)
lines(log10(concentration), k_growth(concentration, 0.5, 0.05), lty = 2, col = "blue", lwd = 2)
lines(log10(concentration), k_growth(concentration, 0.05, 0.055), lty = 2, col = "orange", lwd = 2)
lines(log10(concentration), k_growth(concentration, 0.02216, 0.04786), lty = 2, col = "green3", lwd = 2

abline(v = -1, lty = 2)}
```

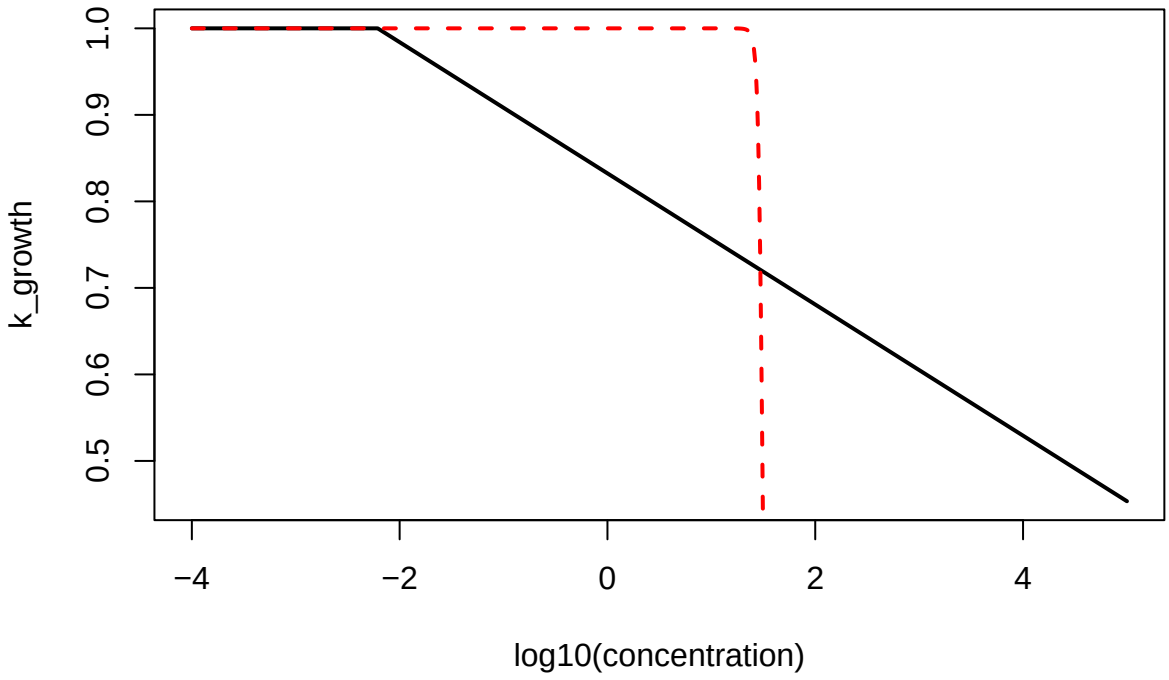



```
# Green curve
nls(dataset$growth~k_growth(dataset$Concentration, EC50, beta_EC), data = dataset, start = c(EC50 = 0.05
```

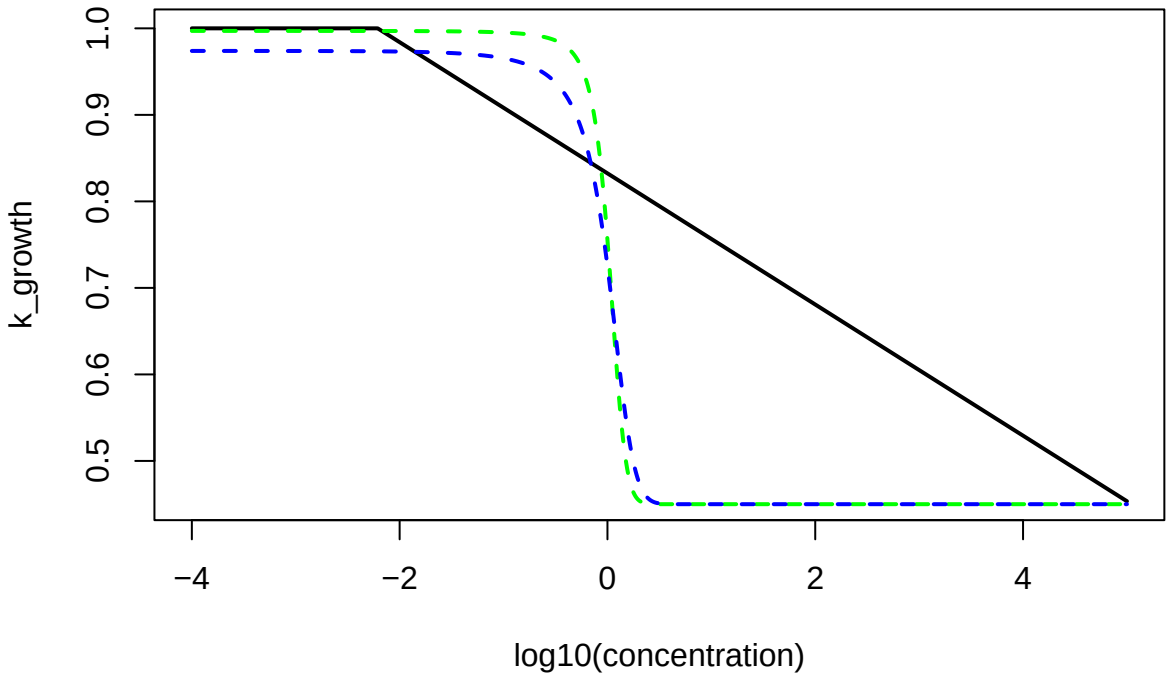
```
## Nonlinear regression model
##   model: dataset$growth ~ k_growth(dataset$Concentration, EC50, beta_EC)
##    data: dataset
##    EC50 beta_EC
## 0.02216 0.04786
##  residual sum-of-squares: 0.08459
##
## Number of iterations to convergence: 5
## Achieved convergence tolerance: 8.312e-06
```

Fit equation 12

```
# Manual adjustement
{plot(log10(concentration), 1-growth/100, ylab = "k_growth", type = "l", lwd = 2)
lines(log10(concentration), k_growth2(concentration, k = 1,  0.7, 31 ), lty = 2, col = "red", lwd = 2)}
```




```
# Best fit
{plot(log10(concentration), 1-growth/100, ylab = "k_growth", type = "l", lwd = 2)
lines(log10(concentration), k_growth2(concentration, k =  0.55,  5, 1.04456 ), lty = 2, col = "green1",
lines(log10(concentration), k_growth2(concentration,k =  0.55,  3, 1 ), lty = 2, col = "blue", lwd = 2)
```



The model does not converge

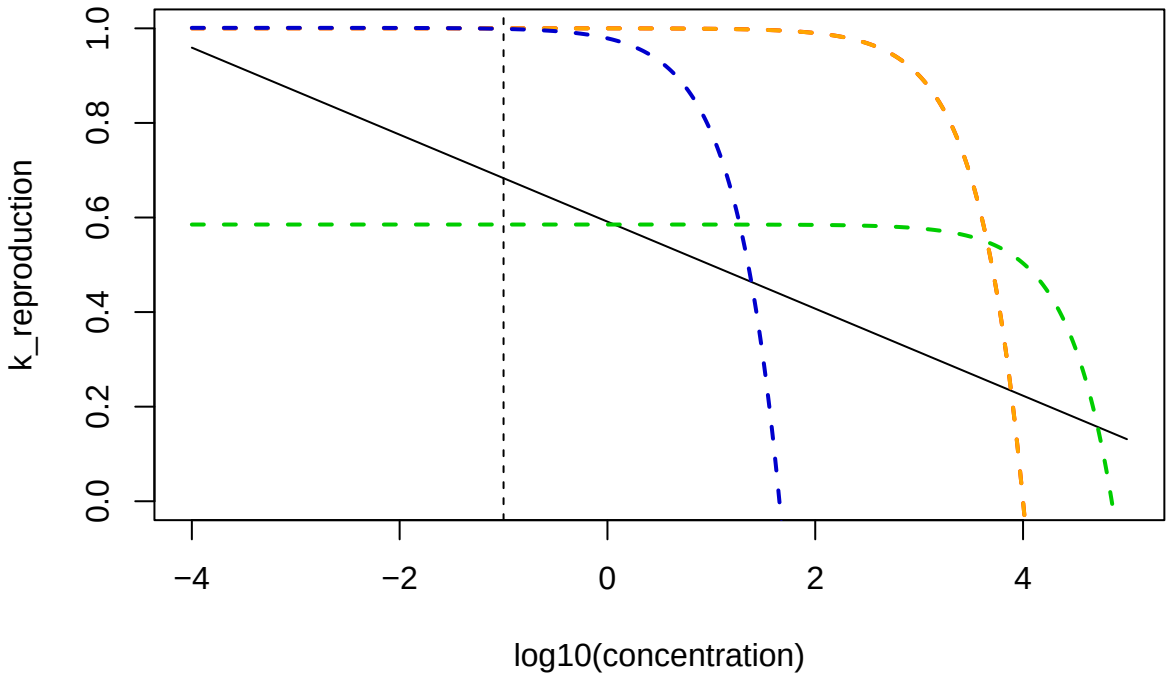
```
#nls(growth~k_growth2(Concentration, k, c_range, c_mid), data = dataset, start = c(k=0.55, c_range =  5
```

Reproduction

Fit equation 13 to data

```
{plot(log10(concentration), 1-reproduction/100, ylab = "k_reproduction", type = "l", ylim = c(0,1))
  # (C_bio, w_r, c_chronic)
lines(log10(concentration), k_repro(concentration, 0.0001, 1), lty = 2, col = "red", lwd = 2)
lines(log10(concentration), k_repro(concentration, 0.0001, 0), lty = 2, col = "orange", lwd = 2)
lines(log10(concentration), k_repro(concentration, 0.02216, 0.04786), lty = 2, col = "blue3", lwd = 2)
lines(log10(concentration), k_repro(concentration,  8.195e-06, -5.062e+04), lty = 2, col = "green3", lw

abline(v = -1, lty = 2)}
```



The slope is "fixed". The second parameter (c_chronic) does not seem to impact a lot the curve. Green is the best curve.

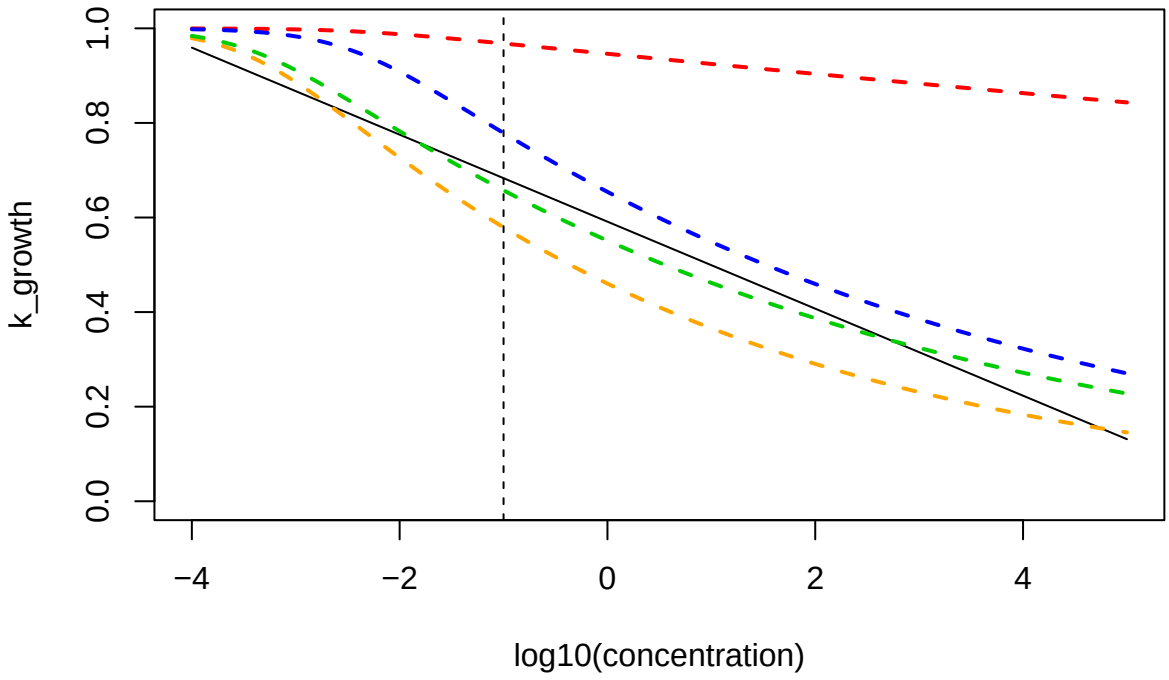
```
nls(reproduction~k_repro(Concentration, w_r, c_chronic), data = dataset, start = c(w_r = 0.02216, c_chr
```

```
## Nonlinear regression model
##   model: reproduction ~ k_repro(Concentration, w_r, c_chronic)
##    data: dataset
##        w_r  c_chronic
##  8.195e-06 -5.062e+04
##  residual sum-of-squares: 38.11
##
## Number of iterations to convergence: 3
## Achieved convergence tolerance: 1.875e-09
```

Fit equation 11 to repro data as a test

```
{plot(log10(concentration), 1-reproduction/100, ylab = "k_growth", type = "l", ylim = c(0,1))
lines(log10(concentration), k_growth(concentration, 0.004, 0.01), lty = 2, col = "red", lwd = 2)
lines(log10(concentration), k_growth(concentration, 0.004, 0.076790), lty = 2, col = "blue", lwd = 2)
lines(log10(concentration), k_growth(concentration, 0.000427, 0.1), lty = 2, col = "orange", lwd = 2)
lines(log10(concentration), k_growth(concentration, 0.000427, 0.076790), lty = 2, col = "green3", lwd =

abline(v = -1, lty = 2)}
```



Green curve = best fit. This equation works better on reproduction data.

```
# Green curve
nls(dataset$reproduction~k_growth(dataset$Concentration, EC50, beta_EC), data = dataset, start = c(EC50
```

```
## Nonlinear regression model
##   model: dataset$reproduction ~ k_growth(dataset$Concentration, EC50,     beta_EC)
##    data: dataset
##     EC50  beta_EC
## 0.000427 0.076790
##  residual sum-of-squares: 1.3
##
## Number of iterations to convergence: 9
## Achieved convergence tolerance: 3.015e-06
```

Mortality

```
#TODO
```

‘‘‘